



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and
Information Technology

EPE321

PROJECT PROPOSAL

CHESS VARIATION:LOS ALAMOS

Name and Surname	Student Number	Signature	% Contribution
E. Mearns	21501212		100

By signing this assignment I confirm that I have read and am aware of the University of Pretoria's policy on academic dishonesty and plagiarism and I declare that the work submitted in this assignment is my own as delimited by the mentioned policies. I explicitly declare that no parts of this assignment have been copied from current or previous students' work or any other sources (including the internet), whether copyrighted or not. I understand that I will be subjected to disciplinary actions should it be found that the work I submit here does not comply with the said policies.

August 11, 2025

1 Problem Statement

Chess is easy to learn but difficult to master. I would like to help bridge this gap and introduce more people to the game of chess by creating a Los Alamos chess platform. Students at the University of Pretoria who want to play chess frequently with class mates or other students attending the university can't easily find opponents at their level, join proper tournaments, or try different chess variations. This chess platform will aim to fix these issues and provide special features for university communities.

Los Alamos is a chess variation that is played on a 6X6 board with simpler rules(no bishops, castling, or en passant moves). It can be a great stepping stone in learning how to play chess and improving your game. The platform can also be used to host tournaments and help students make new friends.

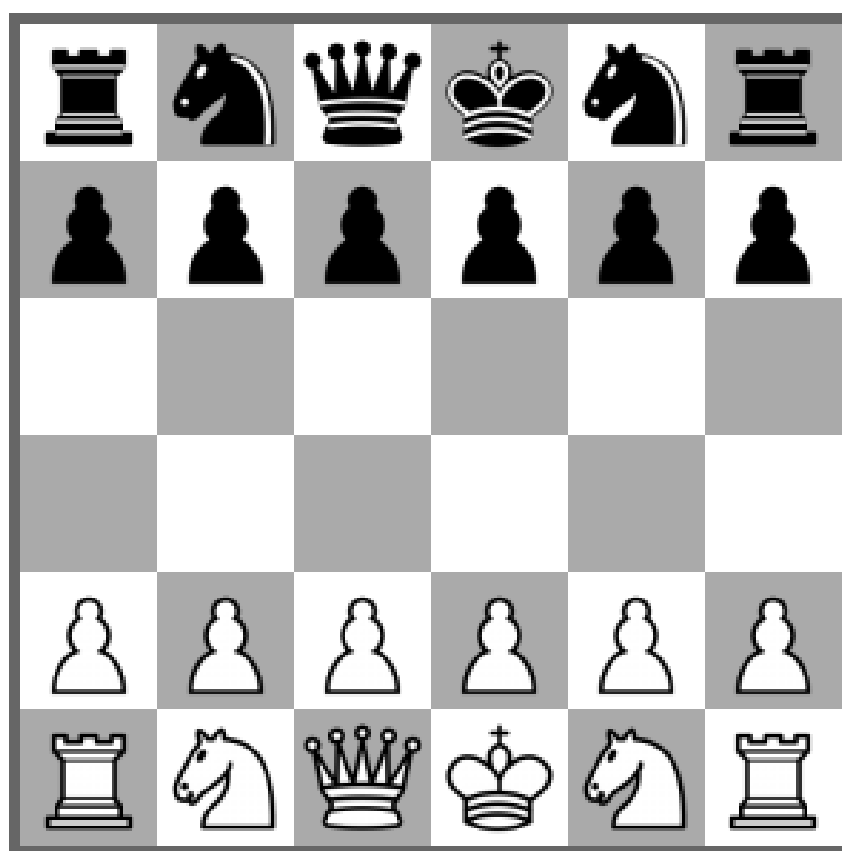


Figure 1: Los Alamos Chess Board (Source: greenchess.net)

2 User Stories (Prioritized)

2.1 High Priority (Must Have)

1. **As a new user**, I want to create an account and set up my profile so I can join the chess community.
2. **As a player**, I want to learn Los Alamos Chess rules through an interactive tutorial so I can understand how to play.

3. **As a player**, I want to play games against other bots or users in real-time so I can practice and enjoy the game.
4. **As a player**, I want to see my game history and statistics so I can track my improvement.
5. **As a tournament organizer**, I want to create and manage tournaments so I can run competitions for the community.

2.2 Medium Priority (Should Have)

1. **As a player**, I want a rating system that matches me with similarly skilled opponents so games are fair and fun.
2. **As a tournament participant**, I want to register for tournaments and see brackets so I can compete properly.
3. **As a user**, I want to view leaderboards and community stats so I can see how I compare.
4. **As a player**, I want to analyze finished games move-by-move so I can learn from mistakes.
5. **As a user**, I want to chat with other players during games so I can make connections.

2.3 Low Priority (Could Have)

1. **As a player**, I want to play against an AI opponent with adjustable difficulty so I can practice anytime.
2. **As a user**, I want to join chess clubs within the platform so I can meet players with similar interests.
3. **As a player**, I want to save and share interesting game positions so I can discuss strategies.

3 Our Development Approach

We will use Agile methods to build the platform. This will feature:

Sprint Cycles: We will present and updated version of the project every two weeks with new features that can be tested. This helps us stay within the scope of the project and understand our clients needs.

Client Involvement: We'll have weekly meetings to plan priorities and present progress. Prototypes will be made regularly available to the clients to test new features.

Real User Testing: Actual students will try out the platform prototypes as we build them. This way we can get active feedback and tailor the design towards the end users.

4 Questions We Need to Answer

Before we start building, we need to decide:

1. How users will log in and make friends (university accounts or new ones?)
2. What tournament formats to use and how often to hold them?
3. How to group players by skill level?
4. Who will manage the platform and its rules?
5. What privacy settings to include(POPI)?

5 Managing Expectations

The user stories contain an ambitious number of features and it is important to know our limits here is what we think is not possible.

5.1 Unreasonable: “Grandmaster-level AI BOTS”

Why it’s hard: Making smart computer bots would take too much time and is outside the scope of the project as the platform is aimed at beginners.

Alternative: We’ll create simpler AI with 3-4 difficulty levels that’s still good for practice.

5.2 Unreasonable: “Support for multiple chess variants”

Why it’s hard: Each different chess type needs lots of special programming.

Alternative: We will keep it simple with one variation Los Alamos.

5.3 Potentially Unreasonable: “Advanced stats and AI coaching and a game history”

Why it’s hard: This needs special skills and lots of time and data we may not have the storage for.

Alternative: We’ll include basic stats and links to good chess learning websites as well as a history of the users previous five games.

5.4 Unreasonable: “24/7 live support”

Why it’s hard: Live support would just be impossible and would be a waste of resources on a smaller project like this.

Alternative: The hope is that the community runs and looks after the platform. This may be naive but other than bug fixes we hope to have a very hands off approach.